1. a) Assume that there is an imaginary node 0, with edges (0,1), and (4,0).
And there is no capacity constraint on those two edges. Let's denote Δ as the flow from 0 to 1. And x_{ij} is the amount of flow across edge (i,j).

∴ The linear problem is:

$$\text{maximize} \quad \Delta$$

$$\text{s.t.} \quad \sum_{j:(j,i)\in E} x_{ji} - \sum_{j:(i,j)\in E} x_{ij} = 0, \quad \forall i \neq 1,4$$

$$\sum_{j:(j,1)\in E} x_{j1} - \sum_{j:(1,j)\in E} x_{1j} + \Delta = 0$$

$$\sum_{j:(j,4)\in E} x_{j4} - \sum_{j:(4,j)\in E} x_{4j} - \Delta = 0$$

$$x_{ij} \le C_{ij}, \quad \forall (i,j) \in E$$

$$x_{ij} \ge 0, \quad \forall (i,j) \in E$$

where $E = \{(1,2), (1,3), (2,3), (2,4), (3,4),\}$

$C_{12}=8,\ C_{13}=7,\ C_{23}=2,\ C_{24}=4,\ C_{34}=12$

Using matlab, we get the result:

```
Maximum Flow: 13
Flow allocation:
{'v1': {'v2': 6, 'v3': 7}, 'v2': {'v3': 2, 'v4': 4}, 'v3': {'v4': 9}, 'v4': {}}
```

The code is attached.

b) The dual problem:

$$\text{minimize} \quad \sum_{(i,j)\in E} C_{ij} z_{ij}, \quad \forall (i,j) \in E$$

$$\text{s.t.} \quad z_{ij} \ge y_i - y_j$$

$$y_s - y_t = 1$$

$$z_{ij} \ge 0$$

From the result we get in (1), we can get the optimal dual value is 13.

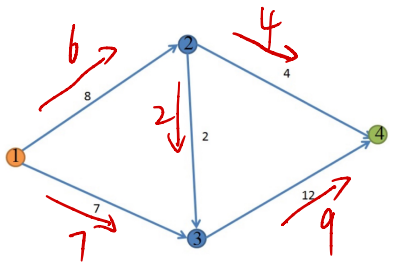
$\because f_{12}=6<C_{12}, \ \therefore z_{12}=0; \ \because f_{34}=9<C_{34}, \ \therefore z_{34}=0;$

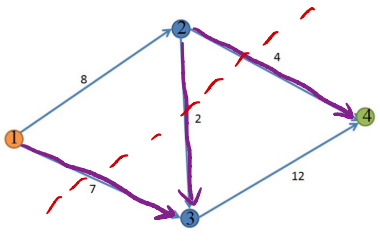
$\because f_{13}=C_{13}, \ f_{23}=C_{23}, \ f_{24}=C_{24}, \ \therefore z_{13}=z_{23}=z_{24}=1$

∴ The dual optimal solution is:

$$\{z_{12}=0,\ z_{13}=1,\ z_{23}=1,\ z_{24}=1,\ z_{34}=0,\ y_1=1,\ y_2=1,\ y_3=0,\ y_4=0\}$$

c) maximum flow: minimum cut:





2. a) The optimal solution is $\{x_1=5, x_2=15\}$

And the optimal value of the problem "minimizing $-5x_1-10x_2$" is -175

therefore the optimal value of the original problem is 175.

b) $AB^{-1}=\begin{bmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ -\frac{2}{3} & 1 & -\frac{10}{3} \\ 0 & 0 & 1 \end{bmatrix}$

$\therefore \begin{bmatrix} 15 \\ 10 \\ 5 \end{bmatrix} + \begin{bmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ -\frac{2}{3} & 1 & -\frac{10}{3} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \lambda \\ 0 \\ 0 \end{bmatrix} \geqslant \vec{0}$

$\therefore \begin{bmatrix} 15 \\ 10 \\ 5 \end{bmatrix} + \begin{bmatrix} \frac{1}{3}\lambda \\ -\frac{2}{3}\lambda \\ 0 \end{bmatrix} \geqslant \vec{0}$

$\therefore -45 \leq \lambda \leq 15$

c) $\therefore \lambda = 10 < 15$

$\therefore$ The new optimal solution is $\begin{bmatrix} 15 \\ 10 \\ 5 \end{bmatrix} + \begin{bmatrix} \frac{1}{3}\cdot 15 \\ -\frac{2}{3}\cdot 15 \\ 0\cdot 15 \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \\ 5 \end{bmatrix}$

$\therefore \{x_1=5, x_2=20\}$, and the optimal value is $5\cdot 5+10\cdot 20=225$

d) $A_N=\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$, $\therefore A_B^{-1}A_N=\begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ -\frac{2}{3} & -\frac{10}{3} \\ 0 & 1 \end{bmatrix}$

$\vec{r}_N^{\,T}=\left[\frac{10}{3}, \frac{5}{3}\right]$

$\therefore \left[\frac{10}{3}, \frac{5}{3}\right] - \lambda[10, 0, 5]\cdot\begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ -\frac{2}{3} & -\frac{10}{3} \\ 0 & 1 \end{bmatrix} \geqslant \vec{0}$

$\therefore \lambda \leq 1$